Groups, Arrays, and Products
Answer Key

Grades 2–3 Operations and Algebraic Thinking

Quick Answers

1. 20	4. 28, 28	7. 24	10. 24
2. 18	5. 18	8. 30	11. 30
3. 21	6. 6	9. 7	12. 4, 8

Curriculum

Level: Grades 2–3 Operations and Algebraic Thinking
3.OA.A.3 / 4.OA.A.2 – *problems 1, 2, 3, 4, 5, 7, 8, 10, 11*
3.OA.C.7 – *problems 4, 6, 9, 12*
Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

10. *If they got 10: added the two numbers instead of making 4 groups of 6*

1. Ana's stickers: $5+5+5+5$.

Step 1. Count how many fives are being added. There are four of them, and every group holds the same amount — that is what makes a multiplication possible.
Step 2. Four groups of five is written 4×5.
Step 3. Count on by fives: $5, 10, 15, 20$.

$$\boxed{4 \times 5 = 20}$$

Listen for: "4 + 5". Ask how many groups there are and how big each group is; the two numbers play different jobs.

2. Array of dots: 3 rows of 6.

Step 1. An array is read as *rows* × *how many in each row*, so count the rows first instead of counting every dot.
Step 2. There are 3 rows, and each row holds 6 dots, so the multiplication is 3×6.
Step 3. Three sixes: $6, 12, 18$.

$$\boxed{3 \times 6 = 18}$$

Teaching note: counting the dots one at a time gives the same answer and is worth allowing once — then ask which way was faster.

3. Write $7 + 7 + 7$ as a multiplication.
Step 1. The same number, 7, is added three times, so this is three equal groups.
Step 2. Three groups of seven is 3×7.
Step 3. $7 + 7 = 14$, and $14 + 7 = 21$.

$$\boxed{3 \times 7 = 21}$$

Listen for: 7×3 written instead. That is the same total, and saying why (three sevens and seven threes both make 21) is a good answer.

4. Array of stars: 4 rows of 7 — two multiplications.
Step 1. Reading across the rows: 4 rows with 7 stars in each row.
Step 2. That is 4×7. Four sevens: $7, 14, 21, 28$.

$$\boxed{4 \times 7 = 28}$$

Step 3. Now read the same picture down the columns. There are 7 columns with 4 stars in each, which is 7×4 — the same array, so the same total.

$$\boxed{7 \times 4 = 28}$$

Why this matters: an array can be read two ways, which is exactly why 4×7 and 7×4 are equal. Not a rule to memorise — a picture to turn.

5. Marco's marbles: 6×3 as repeated addition.
Step 1. 6×3 means *6 groups of 3*, so the number that repeats is 3 and it repeats 6 times.
Step 2. $3 + 3 + 3 + 3 + 3 + 3$.
Step 3. Add in pairs: $3 + 3 = 6$, and there are three pairs, so $6 + 6 + 6 = 18$.

$$\boxed{6 \times 3 = 18}$$

Listen for: $6 + 6 + 6$ (that is 3×6). It has the same total, but it is the other picture — three bags of six, not six bags of three.

6. $4 \times$ _____ $= 24$.
Step 1. The missing number tells how big each of the 4 groups is, so ask: four of what makes 24?
Step 2. Count by fours and watch for 24: $4, 8, 12, 16, 20, 24$ — that is six fours.

$$\boxed{4 \times 6 = 24}$$

Check it: $6 + 6 + 6 + 6 = 24$ ✓ Splitting 24 into four equal groups is the same question, which is why this is also $24 \div 4$.

7. Muffin tray: 6 rows of 4.
Step 1. Rows first: the tray has 6 rows, and each row holds 4 muffins, so the multiplication is 6×4.
Step 2. Count by fours six times: $4, 8, 12, 16, 20, 24$.

$$\boxed{6 \times 4 = 24}$$

Teaching note: a student who knows $4 \times 6 = 24$ may use it here — turning the tray sideways is a legitimate strategy, not a shortcut to discourage.

8. Nia's jumps: $6 + 6 + 6 + 6 + 6$.
Step 1. Count the jumps, not the sixes on the page one by one: there are 5 jumps, and every jump is 6 long.
Step 2. Five groups of six is 5×6.
Step 3. Count by sixes: $6, 12, 18, 24, 30$.

$$\boxed{5 \times 6 = 30}$$

Listen for: landing on 36 — one jump too many. Touching each 6 while counting fixes it.

9. $\underline{\hspace{2em}} \times 5 = 35$.

Step 1. Here the missing number is how many groups there are, and each group holds 5.
Step 2. Count by fives until 35: $5, 10, 15, 20, 25, 30, 35$ — that took seven fives.

$$\boxed{7 \times 5 = 35}$$

Sentence: seven groups of 5 makes 35.

10. Ben wrote $4 + 6 = 10$ for 4×6.

Step 1. $\times$ and $+$ ask different questions. $4 + 6$ puts two piles together once; 4×6 makes *4 groups* that each hold 6.
Step 2. As a repeated addition: $6 + 6 + 6 + 6$.
Step 3. $6 + 6 = 12$, and $12 + 12 = 24$.

$$\boxed{4 \times 6 = 24}$$

What Ben did wrong: he added the two numbers instead of making four groups of six, which is why his answer, 10, is so much smaller than the real product.

11. 4 rows of 6, then one more row.

Step 1. Adding a row does not change how many tiles sit in a row, so the row size stays 6 and only the number of rows changes.
Step 2. (a) 4 rows and one more row makes 5 rows.
Step 3. (b) 5 rows of 6 is $5 \times 6 = 30$ tiles.

$$\boxed{5 \times 6 = 30}$$

Another good route: the old array held $4 \times 6 = 24$ tiles, and the new row adds 6 more: $24 + 6 = 30$ ✓

12. Challenge: Kai's 32 tiles.

Step 1. (a) Each row holds 8 tiles, so ask how many eights make 32: $8, 16, 24, 32$ — four eights.

$$\boxed{\text{4 rows}}$$

Step 2. (b) Four rows of eight, written as repeated addition: $8 + 8 + 8 + 8 = 32$.
Step 3. (c) Now each row holds only 4 tiles, so the rows must be shorter and there must be more of them. Count by fours: $4, 8, 12, 16, 20, 24, 28, 32$ — eight fours.

$$\boxed{\text{8 rows}}$$

Worth pointing out: the same 32 tiles made 4 rows of 8 and then 8 rows of 4. Halving the row size doubled the number of rows.